

KAZACHINSK, Russian Federation

WMO: 303370

Lat: 56.280N

Lon: 107.568E

Elev: 1171

StdP: 14.08

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -47.0	(c) -42.4	(d) -50.2	(e) 0.3	(f) -45.6	(g) -46.2	(h) 0.4	(i) -41.5	(j) 15.8	(k) 3.1	(l) 13.8	(m) 2.6	(n) 0.3	(o) 210	(p) 0.654

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.7	84.6	62.9	81.0	62.1	77.5	61.0	66.9	76.2	65.2	75.6	63.4	73.6	4.9	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.1	93.5	69.9	61.7	85.8	68.5	59.6	79.7	66.7	32.3	75.4	30.9	75.8	29.5	74.0	75.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.1	12.1	10.4	-50.8	91.0	4.8	2.3	-54.3	92.6	-57.1	94.0	-59.8	95.3	-63.3	96.9		
			-49.6	70.1	3.6	2.3	-52.1	71.8	-54.2	73.1	-56.3	74.4	-58.9	76.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.1	-14.0	-4.5	13.1	31.8	45.1	59.2	64.2	59.4	45.1	29.3	5.3	-10.3		
	DBStd	29.54	16.11	14.16	13.14	8.81	7.02	7.03	4.99	5.62	7.45	9.53	16.52	17.09		
	HDD50	9439	1985	1525	1143	548	186	9	0	4	186	643	1342	1868		
	HDD65	13906	2450	1945	1608	997	616	198	76	187	596	1108	1792	2333		
	CDD50	1099	0	0	0	2	36	285	442	295	39	0	0	0		
	CDD65	90	0	0	0	0	1	24	52	13	0	0	0	0		
	CDH74	1560	0	0	0	1	36	573	692	242	16	0	0	0		
	CDH80	399	0	0	0	0	7	158	189	45	0	0	0	0		
	WSAvg	3.6	2.5	2.8	3.9	4.7	5.1	3.9	3.3	3.3	3.7	4.0	3.5	2.6		
	PrecAvg	16.2	0.8	0.6	0.4	0.6	1.3	2.2	2.8	2.8	1.9	1.1	1.0	0.9		
	PrecMax	25.4	1.7	1.2	1.3	1.7	2.8	3.7	7.7	5.6	3.4	2.0	2.0	1.7		
(15) Precipitation	PrecMin	12.6	0.2	0.2	0.0	0.1	0.3	0.7	0.5	0.6	0.1	0.0	0.3	0.3		
	PrecStd	2.8	0.3	0.3	0.3	0.4	0.6	0.9	1.7	1.1	0.7	0.5	0.4	0.4		
	DB	26.5	33.0	49.2	64.1	80.8	88.9	90.0	84.5	76.5	57.2	40.3	29.6			
	MCWB	24.5	27.8	37.1	45.9	57.2	63.4	65.0	64.3	59.0	44.4	35.0	26.8			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	19.5	27.9	42.5	57.0	72.9	84.5	85.3	80.1	68.5	50.8	35.2	21.3			
	2% MCWB	17.9	24.5	33.3	42.1	52.1	61.5	64.5	62.8	54.9	41.2	31.5	19.6			
	5% DB	13.4	22.0	37.4	51.3	67.1	80.9	81.0	76.1	62.3	45.6	31.8	18.0			
	5% MCWB	12.0	19.9	30.4	39.8	49.4	60.7	63.3	61.0	51.7	39.2	29.1	16.8			
	10% DB	8.6	16.9	33.3	46.6	61.6	76.7	77.6	72.0	57.4	41.4	28.4	13.9			
	10% MCWB	7.4	15.0	28.0	37.1	46.8	59.0	62.6	59.9	49.5	36.4	25.9	12.9			
	0.4% WB	24.6	28.0	37.7	47.2	58.2	67.3	69.8	67.7	60.2	46.5	35.5	27.0			
	0.4% MCDB	26.5	33.2	48.3	59.9	77.6	77.1	79.4	76.7	72.6	55.0	38.0	29.8			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	18.0	24.3	34.0	43.2	54.3	64.9	67.8	65.1	56.6	42.6	31.8	19.7			
	2% MCDB	19.6	27.4	41.2	54.7	68.4	78.6	76.3	75.2	65.5	48.7	34.5	21.1			
	5% WB	12.2	20.0	30.9	40.4	51.0	62.9	66.3	63.1	53.6	39.5	29.3	16.7			
	5% MCDB	13.3	21.9	36.9	50.0	63.9	75.4	75.3	71.9	60.0	45.0	32.1	18.0			
	10% WB	7.6	15.1	28.1	37.7	48.1	60.8	64.6	61.2	50.7	36.8	26.2	12.9			
	10% MCDB	8.6	16.8	33.2	46.0	60.1	73.8	74.7	69.7	56.3	41.0	28.4	13.8			
	MDBR	17.6	24.8	29.0	22.7	23.0	24.9	21.7	21.0	18.6	15.6	16.9	16.7			
	5% DB	16.8	24.3	28.9	29.7	35.0	35.4	31.8	30.6	28.6	24.1	15.6	16.3			
(36) Mean Daily Temperature Range	5% WB	15.7	21.1	20.7	16.4	16.4	14.1	12.7	13.5	14.9	14.7	12.7	15.4			
	MCDBR	16.6	23.5	27.7	26.8	30.6	28.3	23.2	24.5	25.2	21.4	14.8	16.0			
	MCWBR	15.5	20.5	20.2	15.7	15.4	12.7	11.3	12.3	13.8	13.7	12.3	15.0			
	taub	0.269	0.301	0.324	0.357	0.402	0.445	0.465	0.406	0.343	0.316	0.281	0.259			
(40) Clear-Sky Solar Irradiance	taud	2.022	2.016	2.051	2.160	2.211	2.172	2.142	2.299	2.440	2.362	2.098	1.987			
	Ebn at Noon	189	231	260	271	265	253	245	254	258	233	183	156			
	Edh at Noon	20	31	38	41	41	43	44	35	27	22	19	16			
	RadAvg	259	568	1031	1471	1637	1793	1650	1370	882	467	256	173			
(44) All-Sky Solar Radiation	RadStd	22	38	48	101	121	143	178	104	82	61	30	20			

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page